Chapter 8. Comparing Estimators

Criteria, sufficiency, UMVUE, asymptotic efficiency

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8.1 Criteria for comparing estimators

The usual performance measure for estimating $\eta = \eta(\theta)$ is the **mean squared error**

$$MSE(\hat{\eta}, \theta) = E_\theta[\{\hat{\eta}(X_1, \ldots, X_n) - \eta(\theta)\}^2] = \mathrm{Var}_\theta\, \hat{\eta} + \{E_\theta \hat{\eta} - \eta(\theta)\}^2,$$

the variance plus the squared **bias**.

Example 8.1 (A clear winner)

$U(0, \theta)$, $n \geq 2$: $\hat{\theta}^{\mathrm{MLE}} = X_{(n)}$ vs $\hat{\theta}^{\mathrm{MME}} = 2\bar{X}$. With $X_{(n)}/\theta \sim \mathrm{Beta}(n, 1)$,

$$MSE(X_{(n)}, \theta) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad MSE(2\bar{X}, \theta) = 4\,\mathrm{Var}\,\bar{X} = \frac{\theta^2}{3n}.$$

The MLE is better for all θ, n, with MSE of order n^{-2} rather than n^{-1}.

Example 8.2 (No uniform winner)

$N(\theta, 1)$: $\bar{X}$ (MSE $1/n$) vs $\hat{\theta}^0 \equiv 0$ (MSE θ^2), better for $|\theta| < 1/\sqrt{n}$; neither dominates. The "pretest" estimator

$$\hat{\theta}^H = \begin{cases} 0, & |\bar{X}| < 1.96/\sqrt{n}, \\ \bar{X}, & |\bar{X}| \geq 1.96/\sqrt{n}. \end{cases}$$

has MSE $n^{-1}E[(Z + \sqrt{n}\theta)^2 \mathbf{1}\{|Z + \sqrt{n}\theta| \geq 1.96\}] + \theta^2 P(|Z + \sqrt{n}\theta| < 1.96)$, $Z \sim N(0, 1)$, a function of $\sqrt{n}\theta$; $n \cdot MSE$ is below 1 near 0 but about 2.5 at $\sqrt{n}|\theta| \approx 2$, and $\to 1$ for fixed $\theta \neq 0$.

MSEs typically cross, so use criteria free of θ:

- **Maximum MSE**: compare $\max_\theta MSE(\hat{\eta}, \theta)$; an estimator minimizing it is **minimax**.
- **Bayes MSE**: for a weight (prior) $\pi \geq 0$, $r(\pi, \hat{\eta}) = \int_\Omega MSE(\hat{\eta}, \theta)\pi(\theta)\, d\theta$; a minimizer is a **Bayes estimator**.

For positive targets, the relative MSE $E_\theta(\hat{\eta}/\eta - 1)^2$ is also used.

Example 8.3

$N(\mu, \sigma^2)$, $\hat{\sigma}_c^2 = c\sum(X_i - \bar{X})^2$; with $V = \sum(X_i - \bar{X})^2/\sigma^2 \sim \chi^2(n-1)$,

$$E_\theta(\hat{\sigma}_c^2/\sigma^2 - 1)^2 = E(cV - 1)^2 = c^2 \cdot 2(n-1) + \{c(n-1) - 1\}^2,$$

free of θ and minimized at $c = 1/(n+1)$: the minimax relative-MSE estimator is neither the MLE ($c = 1/n$) nor S^2.

Alternatively, restrict the class: $\hat{\eta}$ is **unbiased** if $E_\theta \hat{\eta} = \eta(\theta)$ for all θ, and then $MSE = \mathrm{Var}_\theta\, \hat{\eta}$.

Definition (UMVUE)

$\hat{\eta}^*$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $\eta(\theta)$ if (i) it is unbiased, and (ii) $\mathrm{Var}_\theta\, \hat{\eta}^* \leq \mathrm{Var}_\theta\, \hat{\eta}$ for all θ and every unbiased $\hat{\eta}$.

Theorem 8.1 (Uniqueness)

If the UMVUE of $\eta(\theta)$ has finite variance, it is unique: if $\hat\eta_1, \hat\eta_2$ are both UMVUEs, then $P_\theta(\hat\eta_1 = \hat\eta_2) = 1$ for all θ.

Proof. $\hat\eta_3 = (\hat\eta_1 + \hat\eta_2)/2$ is unbiased, so $\operatorname{Var}\hat\eta_3 \geq \operatorname{Var}\hat\eta_2$:

$$\tfrac{1}{4}\{\operatorname{Var}\hat\eta_1 + 2\operatorname{Cov}(\hat\eta_1, \hat\eta_2) + \operatorname{Var}\hat\eta_2\} \geq \operatorname{Var}\hat\eta_2,$$

so $\operatorname{Cov}(\hat\eta_1, \hat\eta_2) \geq \operatorname{Var}\hat\eta_2$ and $\operatorname{Var}(\hat\eta_1 - \hat\eta_2) \leq 0$; a zero-mean, zero-variance variable is 0 a.s. (Theorem 1.11 in Volume I). ■

8.2 Sufficient statistics

Motivation. Sufficiency captures all information about θ in a few numbers; factorization finds such statistics.

Example 8.4

Two Bernoulli(θ) trials, $Y = X_1 + X_2$. For $Y = 0, 2$ the data are determined; for $Y = 1$

$$P_\theta(X_1 = 0, X_2 = 1 \mid Y = 1) = \frac{(1-\theta)\theta}{2\theta(1-\theta)} = \frac{1}{2} = P_\theta(X_1 = 1, X_2 = 0 \mid Y = 1).$$

The data can be recreated from Y by a coin flip without knowing θ.

Definition (Sufficiency)

A statistic $Y = u(X_1, \ldots, X_n)$ is **sufficient** for $\theta \in \Omega$ if the conditional distribution of $(X_1, \ldots, X_n)$ given $Y = y$ does not depend on θ, i.e. $P_{\theta_1}(\mathbf{X} \in A \mid Y = y) = P_{\theta_2}(\mathbf{X} \in A \mid Y = y)$ for all A, y, θ_1, θ_2.

Example 8.5

n Bernoulli(θ) trials, $Y = \sum X_i \sim B(n, \theta)$:

$$P_\theta(X_1 = x_1, \ldots, X_n = x_n \mid Y = y) = \frac{\theta^y(1-\theta)^{n-y}}{\binom{n}{y}\theta^y(1-\theta)^{n-y}} = \binom{n}{y}^{-1} \quad (\textstyle\sum x_i = y),$$

free of θ: $\sum X_i$ is sufficient.

Theorem 8.2 (Factorization)

$Y = u(\mathbf{X})$ is sufficient for $\theta \in \Omega$ if and only if there are functions k_1, k_2 with

$$\prod_{i=1}^n f(x_i; \theta) = k_1(u(\mathbf{x}), \theta)\, k_2(\mathbf{x}) \qquad \text{for all } \mathbf{x} \text{ and } \theta \in \Omega.$$

Proof. Discrete case.

($\Leftarrow$) For $\mathbf{x}$ with $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X} = \mathbf{x} \mid Y = y) = \frac{P_\theta(\mathbf{X} = \mathbf{x})}{\sum_{\mathbf{z}:u(\mathbf{z})=y} P_\theta(\mathbf{X} = \mathbf{z})} = \frac{k_1(y,\theta)k_2(\mathbf{x})}{k_1(y,\theta)\sum_{\mathbf{z}:u(\mathbf{z})=y} k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z})=y} k_2(\mathbf{z})},$$

free of θ (0 if $u(\mathbf{x}) \neq y$).

($\Rightarrow$) $k_2(\mathbf{x}) := P_\theta(\mathbf{X} = \mathbf{x} \mid Y = u(\mathbf{x}))$ is free of θ; with $k_1(y, \theta) = P_\theta(Y = y)$,

$$P_\theta(\mathbf{X} = \mathbf{x}) = P_\theta(\mathbf{X} = \mathbf{x}, Y = u(\mathbf{x})) = k_2(\mathbf{x})\, k_1(u(\mathbf{x}), \theta).$$

■

Example 8.6

(a) Poisson(θ): $\prod f = e^{-n\theta}\theta^{\sum x_i} \cdot \prod(1/x_i!)$, so $\sum X_i$ is sufficient.

(b) Exp(θ): $\prod f = \theta^{-n} e^{-\sum x_i/\theta} \cdot \prod \mathbf{1}\{x_i > 0\}$, so $\sum X_i$ is sufficient.

(c) Gamma(α, β): $\prod f = \{\Gamma(\alpha)\beta^\alpha\}^{-n} (\prod x_i)^{\alpha-1} e^{-\sum x_i/\beta} \cdot \prod \mathbf{1}\{x_i > 0\}$, so $(\sum X_i, \prod X_i)$ is sufficient for (α, β).

Theorem 8.3 (Exponential families)

If $f(x;\theta) = \exp\{\sum_{j=1}^k g_j(\theta) T_j(x) - A(g(\theta)) + S(x)\}$ for $x \in \mathcal{X}$, with support $\mathcal{X}$ not depending on θ, then $(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i))$ is sufficient for θ.

Proof. $\prod f(x_i;\theta) = \exp\{\sum_j g_j(\theta) \sum_i T_j(x_i) - nA(g(\theta))\} \cdot \exp\{\sum S(x_i)\} \prod \mathbf{1}_{\mathcal{X}}(x_i)$; apply the factorization theorem. ■

Theorem 8.4

If Y is sufficient for θ and $W = g(Y)$ with g one-to-one, then W is sufficient.

Proof. $\{W = w\} = \{Y = g^{-1}(w)\}$, so $P_\theta(\mathbf{X} \in A \mid W = w) = P_\theta(\mathbf{X} \in A \mid Y = g^{-1}(w))$, which does not depend on θ. ■

Example 8.7

(a) $N(\mu, \sigma^2)$, $T = (x, x^2)$: $(\sum X_i, \sum X_i^2)$ and its one-to-one image $(\bar{X}, S^2)$ are sufficient.
(b) Beta(α, 1): $f = \exp\{(\alpha - 1)\log x + \log \alpha\}$ on $(0, 1)$, so $\sum \log X_i$ is sufficient.
(c) Support depending on θ: use indicators. Two-parameter exponential $\frac{1}{\sigma} e^{-(x-\mu)/\sigma} \mathbf{1}\{x \geq \mu\}$:

$$\prod f(x_i; \mu, \sigma) = \sigma^{-n} e^{-(\sum x_i - n\mu)/\sigma} \mathbf{1}\{\min x_i \geq \mu\},$$

so $(\sum X_i, X_{(1)})$ is sufficient; for $U[\theta_1, \theta_2]$, $\prod f = (\theta_2 - \theta_1)^{-n} \mathbf{1}\{\theta_1 \leq x_{(1)}, x_{(n)} \leq \theta_2\}$ gives $(X_{(1)}, X_{(n)})$.

Theorem 8.5

If the MLE $\hat{\theta}$ exists uniquely, it is a function of any sufficient statistic $S = u(\mathbf{X})$.

Proof. By factorization $\hat{\theta}(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}), \theta)$ depends on $\mathbf{x}$ only through $u(\mathbf{x})$. ■

A sufficient unique MLE is then a function of every sufficient statistic: **minimal sufficient**, as in Examples 8.5–8.7.

Theorem 8.6 (Rao–Blackwell)

Let Y be sufficient for θ, $\hat\eta$ an estimator of $\eta(\theta)$ with finite variance, and $\hat\eta^{RB} = E[\hat\eta \mid Y]$. Then $\hat\eta^{RB}$ is a statistic, $E_\theta\hat\eta^{RB} = E_\theta\hat\eta$, and

$$MSE(\hat\eta^{RB},\theta) \le MSE(\hat\eta,\theta) \qquad \text{for all } \theta.$$

Proof. The conditional law of $\mathbf{X}$ given Y is free of θ, so $\hat\eta^{RB}$ is a statistic; $E\hat\eta^{RB} = E\hat\eta$ and

$$\operatorname{Var}\hat\eta = E[\operatorname{Var}(\hat\eta \mid Y)] + \operatorname{Var}(E[\hat\eta \mid Y]) \ge \operatorname{Var}\hat\eta^{RB}.$$

With equal bias, $MSE = \text{Var} + \text{bias}^2$ gives the claim. ■

Example 8.8

$U[0,\theta]$: $Y = X_{(n)}$ is sufficient; improve $2\bar X$ (MSE $\theta^2/(3n)$). Given $X_{(n)} = y$, the other order statistics are those of $n-1$ uniforms on $(0,y)$, so $E(X_{(r)} \mid X_{(n)} = y) = \frac{r}{n}y$, $r \le n-1$:

$$\hat\theta^{RB} = \frac{2}{n}E\Big(\sum_{r=1}^{n} X_{(r)} \,\Big|\, X_{(n)}\Big) = \frac{2}{n}\Big(\sum_{r=1}^{n-1}\frac{r}{n} + 1\Big)X_{(n)} = \frac{n+1}{n}X_{(n)},$$

with $MSE = \int_0^\theta \left(\frac{n+1}{n}x - \theta\right)^2 \frac{nx^{n-1}}{\theta^n}dx = \frac{\theta^2}{n(n+2)} \le \frac{\theta^2}{3n}$.

8.3 Minimum variance unbiased estimation

Motivation. Conditioning on a complete sufficient statistic yields the unique best unbiased estimator.

Definition (Completeness)

Y is **complete** for $\theta \in \Omega$ if $E_\theta g(Y) = 0$ for all θ implies $P_\theta(g(Y) = 0) = 1$ for all θ; complete and sufficient: a **complete sufficient statistic**.

Theorem 8.7 (Lehmann–Scheffé)

Let Y be complete sufficient for $\theta \in \Omega$.

(a) If $\hat\eta_0$ is any unbiased estimator of $\eta(\theta)$, then $E[\hat\eta_0 \mid Y]$ is the UMVUE of $\eta(\theta)$.

(b) If $\delta(Y)$ is unbiased for $\eta(\theta)$, then $\delta(Y)$ is the UMVUE.

Proof. (a) $\hat\eta_0^{RB} = E[\hat\eta_0 \mid Y]$ is unbiased. For any unbiased $\hat\eta_1$, $\operatorname{Var}\hat\eta_1^{RB} \le \operatorname{Var}\hat\eta_1$ (Rao–Blackwell); $\hat\eta_0^{RB} - \hat\eta_1^{RB}$ is a function of Y with mean 0, so it is 0 a.s. by completeness, and $\operatorname{Var}\hat\eta_0^{RB} \le \operatorname{Var}\hat\eta_1$.
(b) Take $\hat\eta_0 = \delta(Y)$. ■

Recipe: find a complete sufficient Y, then an unbiased function of Y, or condition a simple unbiased estimator on Y.

Example 8.9 (Bernoulli)

$Y = \sum X_i \sim B(n,\theta)$. If $\sum_y g(y)\binom{n}{y}\theta^y(1-\theta)^{n-y} = 0$ for all $\theta \in (0,1)$, then with $\lambda = \theta/(1-\theta)$

the polynomial $\sum_y g(y)\binom{n}{y}\lambda^y$ vanishes on $(0,\infty)$, so $g \equiv 0$. So Y is complete sufficient and $\bar{X}$ is the UMVUE.

Example 8.10 (Poisson and exponential)

(a) Poisson: $Y \sim \text{Poisson}(n\theta)$; $\sum_y g(y)\lambda^y/y! = 0$ for all $\lambda > 0$ forces $g \equiv 0$ (power series). $\bar{X}$ is the UMVUE.
(b) Exponential: $Y \sim \text{Gamma}(n, \theta)$; $\int_0^\infty g(y)y^{n-1}e^{-\lambda y}dy = 0$ for all $\lambda > 0$ forces $g(y)y^{n-1} = 0$ (Laplace transform). $\bar{X}$ is the UMVUE.

For exponential families (stated without proof, via Laplace transforms):

Theorem 8.8

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$ satisfy:
(i) the support does not depend on η;
(ii) the parameter space N contains an open box $(a_1, b_1) \times \cdots \times (a_k, b_k)$;
(iii) $\mathbf{c}'\mathbf{T}(X)$ is not constant for $\mathbf{c} \neq \mathbf{0}$. Then $\left(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i)\right)$ is complete sufficient.

Example 8.11

(a) $N(\mu, \sigma^2)$: the natural parameter ranges over $\mathbb{R} \times (-\infty, 0)$, so $(\bar{X}, S^2)$ is complete sufficient, and unbiased: the UMVUEs.
(b) $N(\mu, \mu^2)$, $\mu > 0$: the natural parameter lies on a curve (no open box). Indeed $E_\mu Y_2 = 2n\mu^2$ and $E_\mu Y_1^2 = n(n+1)\mu^2$, so $g = \frac{Y_2}{2n} - \frac{Y_1^2}{n(n+1)} \neq 0$ has mean 0: $(\sum X_i, \sum X_i^2)$ is not complete.
(c) $U[0,\theta]$: $Y = X_{(n)}$ has pdf ny^{n-1}/θ^n; $\int_0^\theta g(y)y^{n-1}dy = 0$ for all θ gives $g(\theta)\theta^{n-1} = 0$. So $\frac{n+1}{n}X_{(n)}$ is the UMVUE (Example 8.8).

Example 8.12 (Bernoulli variance)

$\eta = \theta(1-\theta)$, $n \geq 2$: $X_1(1-X_2)$ is unbiased, and

$$E[X_1(1-X_2) \mid Y = y] = \frac{P(X_1 = 1, X_2 = 0, \sum_{i\geq 3} X_i = y-1)}{P(Y = y)} = \frac{\binom{n-2}{y-1}}{\binom{n}{y}} = \frac{y(n-y)}{n(n-1)}.$$

so the UMVUE is $\frac{n}{n-1}\hat{\theta}(1-\hat{\theta})$, $\hat{\theta} = Y/n$.

Example 8.13 (A function of a Poisson mean)

$\eta = e^{-2\theta}$, Poisson(θ): find δ with $E_\theta\delta(Y) = e^{-2\theta}$, $Y \sim \text{Poisson}(n\theta)$:

$$\sum_{y\geq 0}\delta(y)\frac{e^{-n\theta}(n\theta)^y}{y!} = e^{-2\theta} \iff \sum_y \frac{\delta(y)}{y!}\lambda^y = e^{(1-2/n)\lambda} = \sum_y \frac{(1-2/n)^y}{y!}\lambda^y,$$

$\lambda = n\theta$; comparing coefficients, $\delta(y) = (1-2/n)^y \approx e^{-2\bar{X}}$ (the MLE) for large n. For $n = 1$ the UMVUE $(-1)^{X_1}$ takes the value -1 for an estimand in $(0,1)$: unbiasedness can be absurd.

Example 8.14 (Reliability for a normal mean)

$N(\theta,1)$, $n\ge 2$, $\eta = P_\theta(X_1 > a)$: $\mathbf{1}\{X_1 > a\}$ is unbiased and $\bar X$ complete sufficient, so the UMVUE is $P(X_1 > a \mid \bar X)$. As $X_1 - \bar X \sim N(0, \frac{n-1}{n})$ is independent of $\bar X$ (Theorem 4.5 in Volume I),

$$P(X_1 > a \mid \bar X = y) = P(X_1 - \bar X > a - y) = 1 - \Phi\Big(\sqrt{\tfrac{n}{n-1}}\,(a-y)\Big),$$

giving $1 - \Phi\big(\sqrt{n/(n-1)}\,(a-\bar X)\big)$.

A general source of such independence:

Theorem 8.9 (Basu)

If Y is complete sufficient for θ, and the distribution of a statistic Z does not depend on θ (Z is **ancillary**), then Y and Z are independent.

Proof. Fix B. $P(Z\in B \mid Y)$ is free of θ (sufficiency) and $p_B = P_\theta(Z\in B)$ too (ancillarity); $E_\theta[P(Z\in B\mid Y) - p_B] = 0$ for all θ, so completeness gives $P(Z\in B\mid Y) = p_B$ a.s. ∎

Example 8.15 (Reliability for an exponential mean)

$\mathrm{Exp}(\theta)$, $\eta = e^{-a/\theta}$: the UMVUE is $P(X_1 > a \mid Y)$, $Y = \sum X_i$. With $X_i = \theta Z_i$, $Z_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1)$, $X_1/Y \sim \mathrm{Beta}(1, n-1)$ is ancillary, hence independent of Y (Basu):

$$P(X_1 > a \mid Y = y) = P\Big(\frac{X_1}{Y} > \frac{a}{y}\Big) = \int_{a/y}^1 (n-1)(1-z)^{n-2}dz = \Big(1 - \frac{a}{y}\Big)^{n-1}\mathbf{1}\{y > a\},$$

so the UMVUE is $(1 - \frac{a}{n\bar X})^{n-1}\mathbf{1}\{n\bar X > a\}$, close to the MLE $e^{-a/\bar X}$ for large n.

8.4 Asymptotic comparison

Typically squared bias is $O(n^{-2})$ and variance of order n^{-1}, so for large n compare limiting variances.

Definition (Asymptotic relative efficiency)

If $\sqrt{n}(\hat\theta^i_n - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$, $i = 1, 2$, the **asymptotic relative efficiency** of $\hat\theta^1$ with respect to $\hat\theta^2$ is

$$ARE(\{\hat\theta^1_n\}, \{\hat\theta^2_n\}) = \frac{\sigma_2^2(\theta)}{\sigma_1^2(\theta)}.$$

Above 1, $\hat\theta^1$ is more precise: for a 95% error bound $1.96\sigma_i/\sqrt{n_i} = d$, $n_i \approx (1.96/d)^2\sigma_i^2$, so $n_2/n_1 = ARE$.

Example 8.16 (Location families)

Location model $f(x-\theta)$, f symmetric about 0. By Chapter 5 and Theorem 6.13,

$$\sqrt{n}(\hat\theta^{\text{med}} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(0)^2}\Big), \qquad \sqrt{n}(\bar X - \theta) \xrightarrow{d} N(0, \sigma_f^2), \qquad \sqrt{n}(\hat\theta^{\text{MLE}} - \theta) \xrightarrow{d} N(0, I_f^{-1}),$$

with $\sigma_f^2 = \int x^2 f$, $I_f = \int (f'/f)^2 f$:

	$N(\theta,1)$	$L(\theta,1)$	$DE(\theta,1)$
$1/(4f(0)^2)$ (median)	$\pi/2$	4	1
σ_f^2 (mean)	1	$\pi^2/3$	2
I_f^{-1} (MLE)	1	3	1
$ARE(\text{mean},\text{median})$	$\pi/2$	$12/\pi^2$	$1/2$
$ARE(\text{median},\text{MLE})$	$2/\pi$	$3/4$	1

For logistic data: MLE > mean > median; the median needs about 20% more observations than the mean.

Example 8.17 (Beta$(\alpha,1)$)

$m_1=\alpha/(\alpha+1)$, so $\hat\alpha^{\text{MME}}=\bar X/(1-\bar X)$; with $\sigma^2=\frac{\alpha}{\alpha+2}-\frac{\alpha^2}{(\alpha+1)^2}$ and $\dot g(m_1)=(\alpha+1)^2$ for $g(m)=m/(1-m)$, the delta method gives

$$\sqrt{n}(\hat\alpha^{\text{MME}}-\alpha)\xrightarrow{d}N\Big(0,\frac{\alpha(\alpha+1)^2}{\alpha+2}\Big).$$

With $T=\log x$, the MLE solves $\sum\log X_i+n/\alpha=0$: $\hat\alpha^{\text{MLE}}=-1/\overline{\log X}$, $I(\alpha)=1/\alpha^2$, limit $N(0,\alpha^2)$. Hence

$$ARE(\{\hat\alpha^{\text{MME}}\},\{\hat\alpha^{\text{MLE}}\})=\frac{\alpha^2(\alpha+2)}{\alpha(\alpha+1)^2}=\frac{\alpha(\alpha+2)}{(\alpha+1)^2}<1.$$

Why the MLE wins:

Theorem 8.10 (Information inequality)

Assume the regularity conditions (R0)–(R5) of Chapter 6. For any estimator $\hat\eta_n=\hat\eta_n(X_1,\dots,X_n)$ of a real quantity with finite variance,

$$\operatorname{Var}_\theta(\hat\eta_n)\ge\Big(\frac{\partial}{\partial\theta}E_\theta\hat\eta_n\Big)'[nI(\theta)]^{-1}\Big(\frac{\partial}{\partial\theta}E_\theta\hat\eta_n\Big).$$

Proof. One-dimensional θ: by Cauchy–Schwarz,

$$\operatorname{Var}_\theta(\hat\eta_n)\operatorname{Var}_\theta(\dot l_n(\theta))\ge\{\operatorname{Cov}_\theta(\hat\eta_n,\dot l_n(\theta))\}^2.$$

By Theorem 6.12, $E_\theta\dot l_n(\theta)=0$ and $\operatorname{Var}_\theta\dot l_n(\theta)=nI(\theta)$; by (R4)

$$\operatorname{Cov}_\theta(\hat\eta_n,\dot l_n(\theta))=E_\theta[\hat\eta_n\dot l_n(\theta)]=\int\hat\eta_n(\mathbf{x})\frac{\partial}{\partial\theta}\prod f(x_i;\theta)\,d\mathbf{x}=\frac{\partial}{\partial\theta}E_\theta\hat\eta_n.$$

Substitute. For vector θ, with $\Sigma_{11}=\operatorname{Var}\hat\eta_n$, $\Sigma_{12}=(\partial_\theta E_\theta\hat\eta_n)'$, $\Sigma_{22}=nI(\theta)$,

$$\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}=\operatorname{Var}\left(\hat\eta_n-\Sigma_{12}\Sigma_{22}^{-1}\dot l_n(\theta)\right)\ge 0.$$

■

Corollary 8.11 (Cramér–Rao bound)

If $\hat\theta^{UE}$ is unbiased for $\theta\in\mathbb{R}^k$, then $\mathbf{c}'\operatorname{Var}_\theta(\hat\theta^{UE})\mathbf{c}\ge\mathbf{c}'I(\theta)^{-1}\mathbf{c}/n$ for all $\mathbf{c}$; i.e. $\operatorname{Var}_\theta(\hat\theta^{UE})-I(\theta)^{-1}/n$ is nonnegative definite. For $k=1$, $\operatorname{Var}_\theta(\hat\theta^{UE})\ge 1/\{nI(\theta)\}$.

Proof. Apply Theorem 8.10 to $\hat{\eta} = \mathbf{c}'\hat{\theta}^{UE}$, for which $\partial_\theta E_\theta \hat{\eta} = \mathbf{c}$. ■

Example 8.18

Beta$(\alpha, 1)$: $Y = \sum \log X_i$ is complete sufficient and $-\alpha Y \sim$ Gamma$(n, 1)$, so

$$E\Big[\frac{1}{-Y}\Big] = \frac{\alpha}{n-1}, \qquad \text{Var}\Big[\frac{1}{-Y}\Big] = \frac{\alpha^2}{(n-1)^2(n-2)}.$$

So $\hat{\alpha}^{\text{UMVUE}} = (n-1)/(-Y) = \frac{n-1}{n}\hat{\alpha}^{\text{MLE}}$ has variance $\alpha^2/(n-2)$, above the bound α^2/n but asymptotically equal, with the MLE's limit $N(0, \alpha^2)$.

So $\text{Var}\{\sqrt{n}(\hat{\theta}^{UE} - \theta)\} \geq 1/I(\theta)$, suggesting every asymptotically normal estimator has limiting variance at least $1/I(\theta)$ (**Fisher's conjecture**); false in general, true under extra conditions such as locally uniform convergence. In this sense the MLE is **asymptotically efficient**.

Example 8.19 (Gamma)

Gamma(α, β): the MME (Example 6.1(d)) satisfies

$$\sqrt{n}(\hat{\theta}^{\text{MME}} - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma(\theta)), \qquad \Sigma(\theta) = \begin{pmatrix} 2\alpha(\alpha+1) & -2(\alpha+1)\beta \\ -2(\alpha+1)\beta & (2+3/\alpha)\beta^2 \end{pmatrix},$$

and the MLE (or one-step, Example 6.16) has limit covariance $I(\theta)^{-1}$:

$$ARE(\hat{\alpha}^{\text{MME}}, \hat{\alpha}^{\text{MLE}}) = \frac{1}{2(\alpha+1)\{\alpha\dot{\Psi}(\alpha) - 1\}},$$

$$ARE(\hat{\beta}^{\text{MME}}, \hat{\beta}^{\text{MLE}}) = \frac{\alpha\dot{\Psi}(\alpha)}{(2\alpha+3)\{\alpha\dot{\Psi}(\alpha) - 1\}}.$$

About 0.39 and 0.51 at $\alpha = 1$, increasing to 1 as $\alpha \to \infty$ but always below 1.